MT CLASSES

CLASS 10 · SCIENCE · CHAPTER 7

How do Organisms Reproduce?

Complete Chapter Notes · CBSE 2026-27 · NCERT-aligned

Inside: theory for every NCERT topic · 12 labelled diagrams · comparison tables · flowcharts · all NCERT questions answered · exam practice with answers

Name: _____ Batch: _____ Date: _____

Chapter Snapshot

- **Unit II — World of Living** carries **25 marks** in the 80-mark theory paper (shared with Life Processes, Control and Coordination, and Heredity).
- This is **Chapter 7** of the rationalised NCERT Class 10 Science textbook, which is unchanged for the 2026-27 session.
- **CBSE syllabus points:** asexual and sexual reproduction in plants and animals; reproductive health — need and methods of family planning; safe sex vs HIV/AIDS; child bearing and women's health.

Chapter Map

Section	What it covers
1	Why do organisms reproduce?
2	DNA copying and the importance of variation
3	Modes of reproduction — an overview
4	Asexual reproduction: fission, fragmentation, regeneration, budding, vegetative propagation, spore formation
5	Sexual reproduction — why it matters
6	Sexual reproduction in flowering plants
7	Reproduction in human beings
8	Reproductive health — STDs, HIV/AIDS, contraception, women's health
9	Quick revision — key terms, organism table, diagrams list, memory tricks, common mistakes
10	All NCERT in-text and exercise questions — answered
11	Exam practice — MCQs, Assertion–Reason, short, long and case-based questions

1. Why Do Organisms Reproduce?

Reproduction is the process by which living organisms produce new individuals of their own kind.

- Life processes such as nutrition, respiration and excretion are needed to keep an individual alive. **Reproduction is not** — an organism can live its whole life without ever reproducing.

- However, reproduction is **essential for the species**. Every individual dies one day; only reproduction creates new members, so the species continues generation after generation (**continuity or perpetuation of the species**).
- We notice a kind of organism only because there are **large numbers** of it around us. A single, isolated individual would hardly be noticed. These large numbers are the result of reproduction.
- Offspring look like their parents because they inherit the same **basic body design**, whose instructions are carried in **DNA**.

> **Exam Tip**
>
> A favourite 1-mark and Assertion–Reason idea: *reproduction is not essential for the survival of an individual, but it is essential for the survival of a species.*

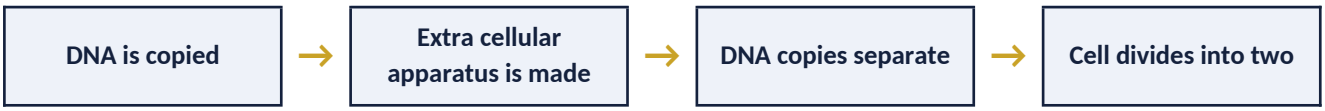
2. Do Organisms Create Exact Copies of Themselves?

2.1 DNA — the blueprint of the body

- The **chromosomes** in the **nucleus** of a cell carry the information for passing on features from parents to offspring. This information is stored in **DNA (deoxyribonucleic acid)**.
- DNA is the **information source for making proteins**. If the information changes, different proteins are made, and the **body design** of the organism can change.
- Therefore, the basic event in reproduction is the **creation of a DNA copy**.

2.2 How does a cell make a copy of itself?

1. The cell uses **chemical reactions** to build a copy of its DNA, so there are now **two copies**.
2. Along with DNA copying, **additional cellular apparatus** (membranes, organelles, cytoplasm) is made, so that each DNA copy has the machinery it needs to work.
3. The two DNA copies **separate**, each taking its own cellular apparatus.
4. The cell **divides into two**. One cell has given rise to two cells.



2.3 Copies are similar, not identical — variation

- No chemical reaction is 100% reliable. So the DNA copies are **similar to the original but not exactly identical**.
- These small copying errors are the source of **variation**.
- Some changes are so drastic that the new DNA cannot work with the cellular apparatus it inherits — **such a new cell simply dies**.
- Other changes are minor. The cell survives, and the offspring show **small variations**.
- Consistent DNA copying is still important: it keeps the **body design** that allows an organism to use its particular **niche** (its place and role in the environment). This links reproduction to the **stability of populations** of a species.

> **Remember**
>
> DNA copying must be **reliable** (so offspring get the parents' body design) but it is **never perfect** (so offspring show variation). Good answers mention both.

2.4 The importance of variation

- Every population lives in a well-defined **niche** and uses its body design to survive there.
- A niche can change drastically for reasons beyond the organisms' control — a rise in temperature, a fall in water level, even a meteorite hit.
- If all members of a population were identical and perfectly suited to the old niche, a drastic change could **wipe out the whole population**.
- If some members carry variations, a few of them may **survive the new conditions**, reproduce and keep the species going.
- **NCERT example:** bacteria living in temperate water. If global warming raises the water temperature, most of them die, but the few **heat-resistant variants survive and multiply**.
- Variation is therefore useful for the **survival of the species over time**, and it is the basis of **evolution**.
- But variation is **not necessarily useful to the individual** that carries it — for that individual it may be useless or even harmful. Its benefit shows up at the population level, when the environment changes.

3. Modes of Reproduction — An Overview

- **Asexual reproduction:** a **single parent** produces new individuals **without forming or fusing gametes**. The offspring are almost exact copies of the parent and of one another (**clones**).
- **Sexual reproduction:** usually **two parents** (male and female) are involved. Special cells called **gametes** (germ cells) are formed and they **fuse** during fertilisation. The offspring show **variation**.

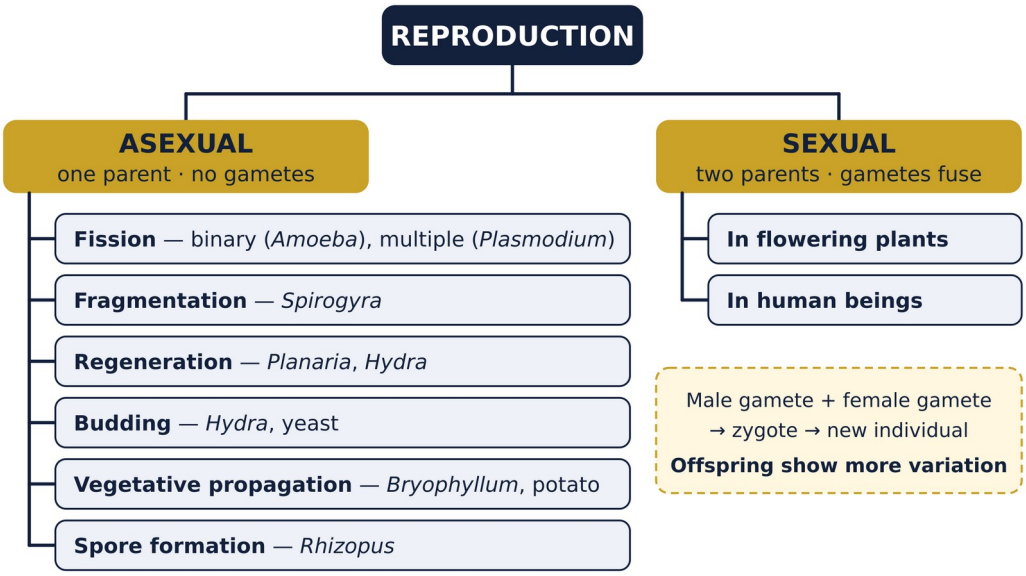


Fig. 1: Modes of reproduction at a glance

Feature	Asexual reproduction	Sexual reproduction
Parents	One	Usually two
Gametes	Not formed	Male and female gametes formed

Feature	Asexual reproduction	Sexual reproduction
Fertilisation	Absent	Present — gametes fuse to form a zygote
Cell division	Only ordinary division of body cells (mitosis)	A special reduction division (meiosis) forms gametes; ordinary division for growth
Variation	Very little — only from DNA-copying errors	Much more — DNA of two parents combines
Speed	Fast; many offspring in a short time	Slower; needs more time and energy
Role in evolution	Small	Large — provides the raw material for evolution
Examples	*Amoeba*, *Hydra*, yeast, *Rhizopus*, *Bryophyllum*	Human beings, most animals, flowering plants

4. Asexual Reproduction — Modes Used by Single Organisms

4.1 Fission

In **fission**, a single-celled organism divides to form two or more new individuals. The parent disappears as it becomes the daughter cells. Fission is common in bacteria and protozoa.

(a) Binary fission

- The parent cell splits into **two equal halves**, and each half becomes a new individual.
- The **nucleus divides first**, and then the **cytoplasm** divides.
- Examples: ***Amoeba***, ***Paramecium***, ***Leishmania*** and many **bacteria**.
- ***Amoeba*** has no fixed shape, so it can split **in any plane**.
- ***Leishmania*** (the parasite that causes **kala-azar**) has a **whip-like structure (flagellum)** at one end. Its binary fission takes place in a **definite orientation** in relation to this structure.

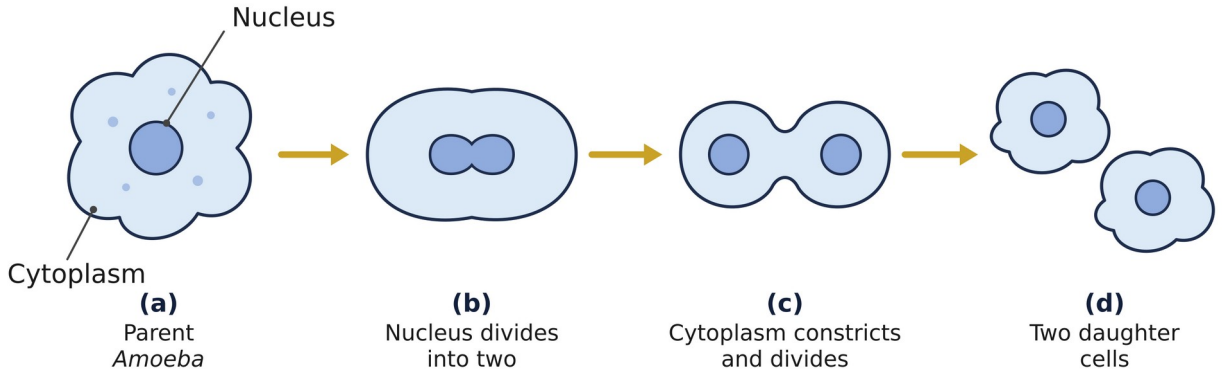


Fig. 2: Binary fission in *Amoeba*

> **Local Connect**
>
> Kala-azar (black fever) is caused by *Leishmania* and spread by the bite of sandflies. Bihar has been one of the states most affected by it — so this textbook example is close to home.

(b) Multiple fission

- The parent cell divides into **many daughter cells at the same time**.

- The **nucleus divides repeatedly** to form many nuclei. Each nucleus gathers a little cytoplasm around itself, and the parent then breaks up to release many daughter cells.
- Example: ***Plasmodium***, the malarial parasite (malaria spreads through the bite of the female *Anopheles* mosquito). *Plasmodium* multiplies this way inside human liver cells and red blood cells.
- ***Amoeba*** can also divide by multiple fission when conditions are unfavourable: it forms a protective wall called a **cyst** and divides into many daughter cells inside it.

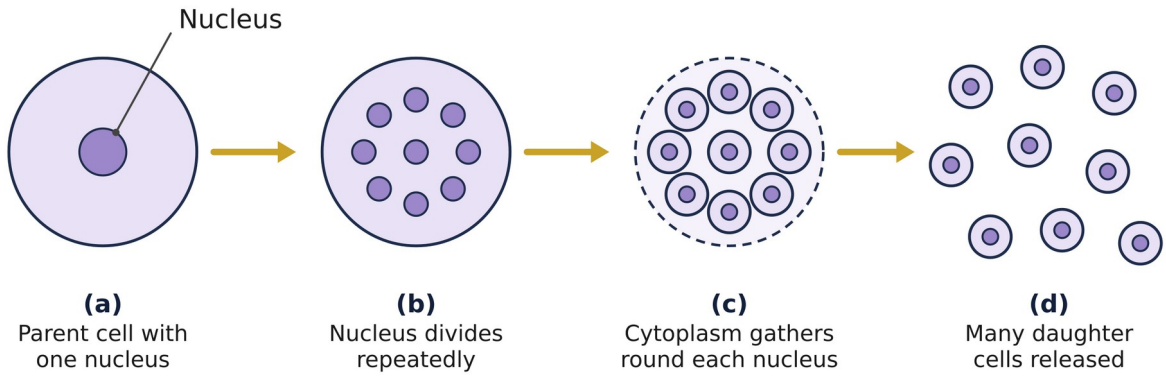


Fig. 3: Multiple fission in *Plasmodium*

Point	Binary fission	Multiple fission
Daughter cells	Two	Many, at the same time
Division of nucleus	Divides once	Divides repeatedly
Division of cytoplasm	Cytoplasm splits into two	Cytoplasm collects around each nucleus
Examples	*Amoeba*, *Paramecium*, *Leishmania*, bacteria	*Plasmodium* (and *Amoeba* inside a cyst)

4.2 Fragmentation

- In multicellular organisms with a **simple body organisation**, the mature body simply **breaks up into smaller pieces (fragments)**, and **each fragment grows into a new individual**.
- Example: ***Spirogyra***, a green, thread-like (filamentous) alga found in ponds.
- This works for *Spirogyra* because its filament is made of **similar cells** — every fragment can carry out all the life processes.
- Most multicellular organisms cannot do this. They are **not a random collection of cells**: specialised cells form **tissues and organs** placed at definite positions. Simple cell-by-cell division cannot rebuild such a body, so these organisms need more complex ways of reproducing.

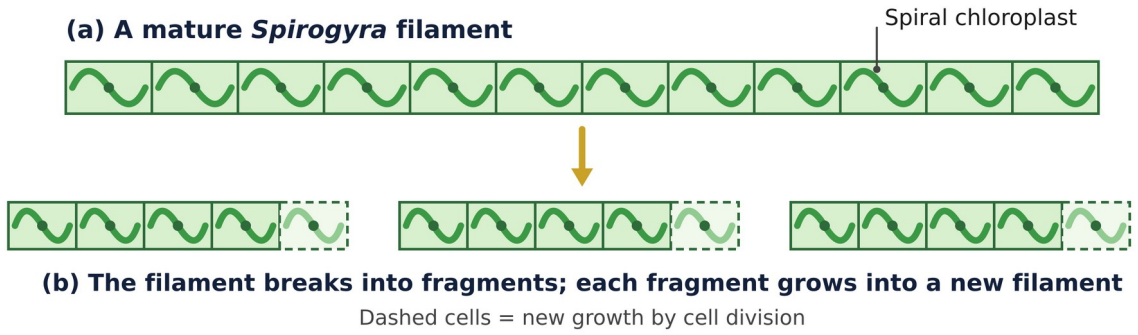


Fig. 4: Fragmentation in *Spirogyra*

NCERT Activity — *Spirogyra*

Collect dark green, thread-like filaments from a pond. Place one or two filaments on a slide with a drop of glycerine, cover with a coverslip and observe under a microscope. All the cells look alike — there are no different tissues.

4.3 Regeneration

- Many fully differentiated organisms can form complete new individuals from their **body parts**. If the body is cut into several pieces, many of the pieces grow into **separate individuals**.
- Examples: ***Planaria*** (a flatworm) and ***Hydra***.
- Regeneration is carried out by **specialised cells**:
  - they **proliferate** (divide again and again) to make a large mass of cells;
  - cells in this mass then **change into different cell types and tissues**;
  - these changes take place in an **organised sequence** called **development**.
- Regeneration is **not the same as reproduction** — most organisms do not normally depend on being cut into pieces in order to reproduce.

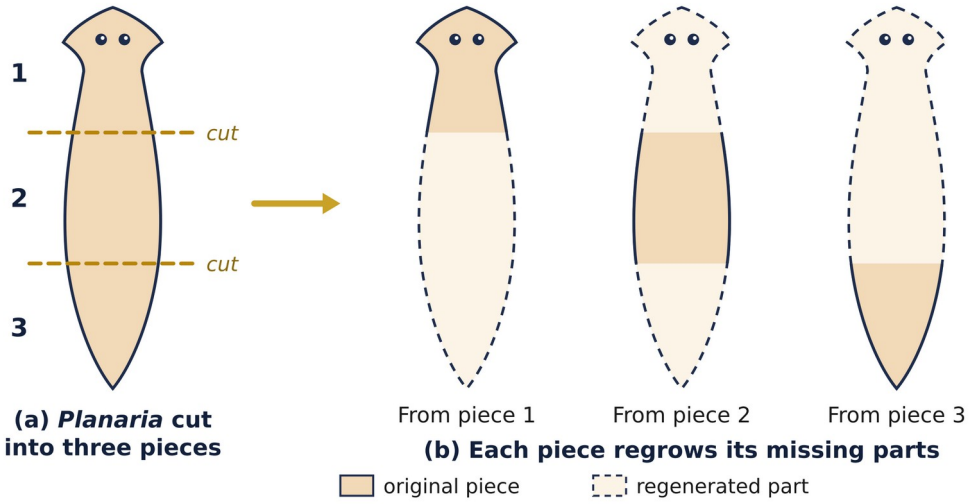


Fig. 5: Regeneration in *Planaria*

Why can't complex organisms regenerate a whole body?

- Their bodies have highly specialised tissues, organs and organ systems that must work together.
- Most of their cells are specialised and have lost the ability to divide and form all the other kinds of cells.
- They do not have enough regenerative cells to rebuild a complete body from a small part.

Complex animals can only repair or regrow small parts — for example, healing of a wound, or a lizard regrowing its tail.

4.4 Budding

- A small **outgrowth (bud)** forms on the parent body at a **specific site** because of repeated cell division there.
- The bud grows into a **tiny individual**. When fully mature, it **detaches** from the parent and lives independently.

- Examples: ***Hydra*** (it uses its regenerative cells to form buds) and **yeast**, a single-celled fungus. In yeast, buds sometimes stay attached for a while and form a **chain**.

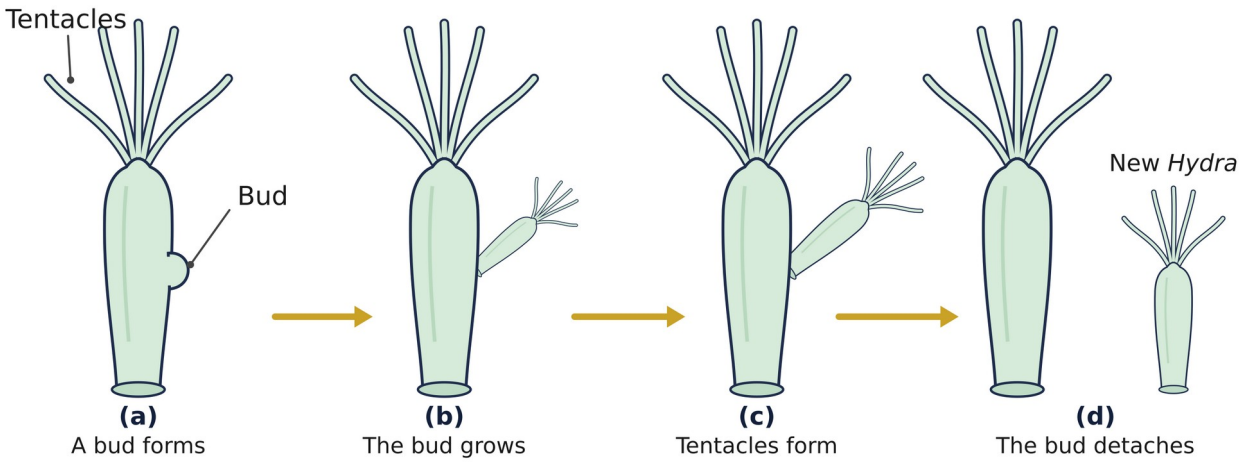


Fig. 6: Budding in *Hydra*

NCERT Activity — Budding in yeast

Dissolve about 10 g of sugar in 100 mL of water. Take 20 mL of this solution in a test tube, add a pinch of yeast granules, plug the tube with cotton and keep it in a warm place. After 1–2 hours, observe a drop under a microscope: yeast cells with small buds (sometimes chains of buds) can be seen.

Point	Binary fission	Budding
Division	Parent splits into two equal halves	A small bud grows out; division is unequal
Parent	Loses its identity	Remains, and can form more buds
Examples	*Amoeba*, *Leishmania*	*Hydra*, yeast

Point	Fragmentation	Regeneration
How it starts	The mature body breaks up into pieces	The body is cut or broken, e.g., by injury
Organisms	Simple multicellular, e.g., *Spirogyra*	Fully differentiated, e.g., *Planaria*, *Hydra*
Nature	A normal mode of reproduction	Mainly repair; not a normal way to reproduce
Process	Fragments simply grow by cell division	Specialised cells proliferate and differentiate in an organised way

4.5 Vegetative Propagation

In many plants, new plants grow from **vegetative parts — root, stem or leaf** — under suitable conditions. No flowers, seeds or gametes are involved.

(a) Natural methods

- **By leaves:** in ***Bryophyllum***, buds form in the **notches along the leaf margin**. When they fall on the soil, they grow into new plants.
- **By stems:** the **potato tuber** is a stem; its 'eyes' are **buds**, and each can grow into a new plant. Ginger (rhizome) and onion (bulb) are other stem examples.
- **By roots:** sweet potato and dahlia — their roots can give rise to new plants.

(b) Artificial methods (used in agriculture and gardening)

- **Cutting:** a piece of stem with buds (nodes) is planted in moist soil; it grows roots and becomes a new plant — e.g., rose, sugarcane, money plant.
- **Layering:** a branch is bent down and covered with soil while still attached to the parent. Once it develops roots, it is cut off — e.g., jasmine, grapevine.
- **Grafting:** a shoot of a desired plant (the **scion**) is joined to a rooted plant (the **stock**) so that both grow as one plant — e.g., mango, rose, citrus.

(c) Advantages

1. Plants that have **lost the capacity to produce seeds** — such as banana, orange, rose and jasmine — can still be propagated.
2. Such plants **bear flowers and fruits earlier** than plants raised from seeds.
3. All the new plants are **genetically similar to the parent**, so desirable features (taste, yield, colour, disease resistance) are preserved.
4. It is a quick, easy and cheap way to raise many plants.

(d) Limitations

- **No variation** — all the plants are alike, so a single disease or change in climate can damage the whole crop.
- In nature, new plants grow close to the parent and **compete** with it for water, light and nutrients.

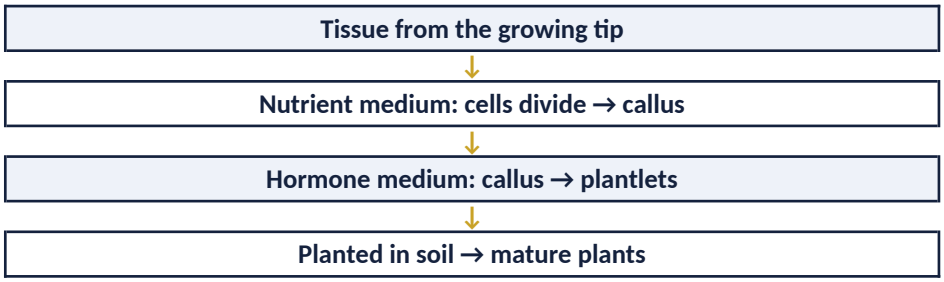
NCERT Activities — Potato and money plant

Potato: Cut a potato into small pieces, some with an 'eye' (bud) and some without. Keep them on wet cotton. Only the pieces with a bud sprout — a bud is needed to form a new plant.

Money plant: Cut some pieces that carry at least one leaf, and other pieces from between two leaves. Keep one end of each in water. Pieces with a leaf (they have a node with a bud) grow roots and new leaves; pieces cut from between two leaves do not.

(e) Tissue culture (micropropagation)

1. A small piece of tissue is taken from the **growing tip** of a plant.
2. It is placed in an **artificial nutrient medium** under sterile conditions. The cells divide rapidly to form a mass of undifferentiated cells called **callus**.
3. The callus is moved to another medium containing **plant hormones** for growth and differentiation; small **plantlets** form.
4. The plantlets are planted in soil, where they grow into mature plants.



- **Advantages:** a large number of plants from a single parent; **disease-free** plants; can be done all year round; useful for plants that do not produce viable seeds.

- **Uses:** commonly used for **ornamental plants** such as orchids, carnation and chrysanthemum, and for crops like **banana**.

4.6 Spore Formation

- Example: ***Rhizopus*** (bread mould), a fungus.
- Its body is made of thread-like structures called **hyphae**, which spread over the bread.
- Tiny **'blob-on-a-stick'** structures rise from the hyphae. Each blob is a **sporangium** (plural: sporangia), and it contains many cells called **spores**.
- Spores have **thick walls** that protect them until they come in contact with a **moist surface**. There they germinate and grow into new *Rhizopus*.
- **Benefits of reproducing through spores:**
  - spores are produced in **very large numbers**;
  - they are **tiny and light**, so air carries them far, and the organism spreads to new places;
  - the **thick wall** lets them survive heat, dryness and other unfavourable conditions until conditions improve.

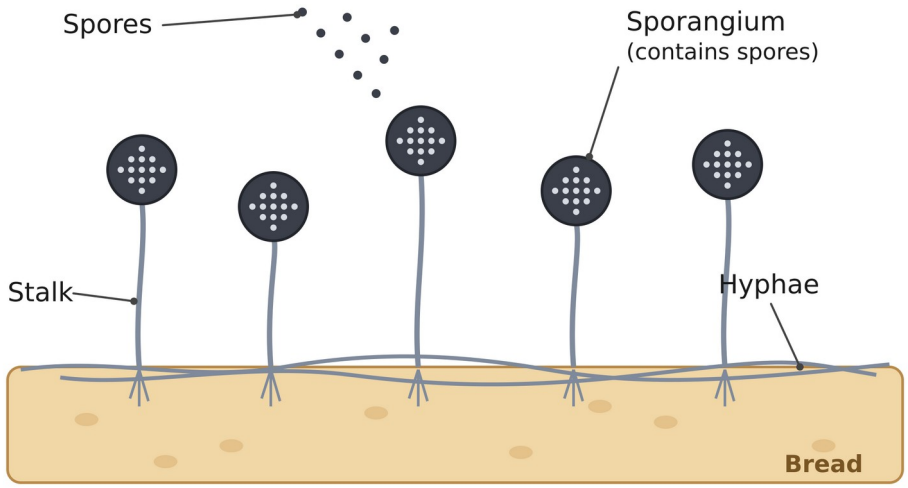


Fig. 7: Spore formation in *Rhizopus*

NCERT Activity — Bread mould

Keep a moist slice of bread in a cool, moist and dark place. Observe its surface with a magnifying glass for a week. First, white cottony threads (hyphae) appear; later, black dots (sporangia) are seen on them.

4.7 Summary of asexual modes

Mode	What happens	Examples
Binary fission	One cell splits into two equal daughter cells	*Amoeba*, *Leishmania*, *Paramecium*, bacteria
Multiple fission	One cell splits into many daughter cells at once	*Plasmodium*
Fragmentation	The mature body breaks into pieces; each piece grows	*Spirogyra*
Regeneration	Cut pieces of the body grow into whole organisms	*Planaria*, *Hydra*
Budding	An outgrowth (bud) grows and separates	*Hydra*, yeast

Mode	What happens	Examples
Vegetative propagation	New plants from root, stem or leaf	*Bryophyllum*, potato, sugarcane, rose
Spore formation	Sporangia release spores that grow into new individuals	*Rhizopus*

5. Sexual Reproduction

5.1 Why the sexual mode of reproduction?

- In asexual reproduction, new variations come only from small errors in DNA copying, so they **build up slowly**.
- In sexual reproduction, the **DNA of two individuals is combined** in the offspring. New combinations appear much faster, so the offspring show **much more variation**.
- More variation means a greater chance that some individuals will survive when the environment changes. This helps the **survival of the species** and provides the raw material for **evolution**.

5.2 The problem of doubling — and its solution

- If two ordinary body cells, each with a full set of DNA, were to fuse, the new cell would have **double the DNA**. The amount would keep doubling every generation, and the cell's DNA-controlled machinery would break down.
- The solution: **germ cells (gametes)** are made in specialised reproductive organs by a special kind of cell division (**meiosis**). Each gamete has **only one set of chromosomes — half the DNA** of a body cell.
- When a male gamete and a female gamete fuse during **fertilisation**, the **zygote** gets the **normal number of chromosomes** back.
- In this way, the chromosome number and the amount of DNA **stay constant** from one generation to the next.

> **Example — Human beings**
>
> Body cells have **46 chromosomes** (23 pairs). The sperm and the egg have **23 chromosomes** each. At fertilisation: 23 + 23 = **46** chromosomes in the zygote.

5.3 The two kinds of gametes

- In simple organisms, the two gametes may look alike.
- In complex organisms they are **specialised**:
  - the **male gamete** is **small and motile** (it can move);
  - the **female gamete** is **larger, non-motile** and contains **stored food**.
- Because different organs produce these gametes, the bodies of males and females in many animals are also different.

5.4 Advantages of sexual reproduction

1. Combining the DNA of two parents produces **more variation**.
2. Variation helps the species **adapt** to changing surroundings, so it is more likely to survive.
3. Variation is the **raw material for evolution**; over long periods it can lead to new species.

4. Useful features of **two different parents** can come together in the offspring.

6. Sexual Reproduction in Flowering Plants

6.1 The flower — the reproductive part of the plant

In flowering plants (angiosperms), the reproductive parts are located in the **flower**. From outside to inside, a typical flower has:

- **Sepals** — green and leaf-like; they protect the flower in the bud stage.
- **Petals** — often brightly coloured; they attract insects for pollination.
- **Stamens** — the **male** reproductive part.
- **Pistil (carpel)** — the **female** reproductive part, found in the **centre** of the flower.
- The flower stalk is the **pedicel**; its swollen tip, on which all the parts are attached, is the **thalamus (receptacle)**.

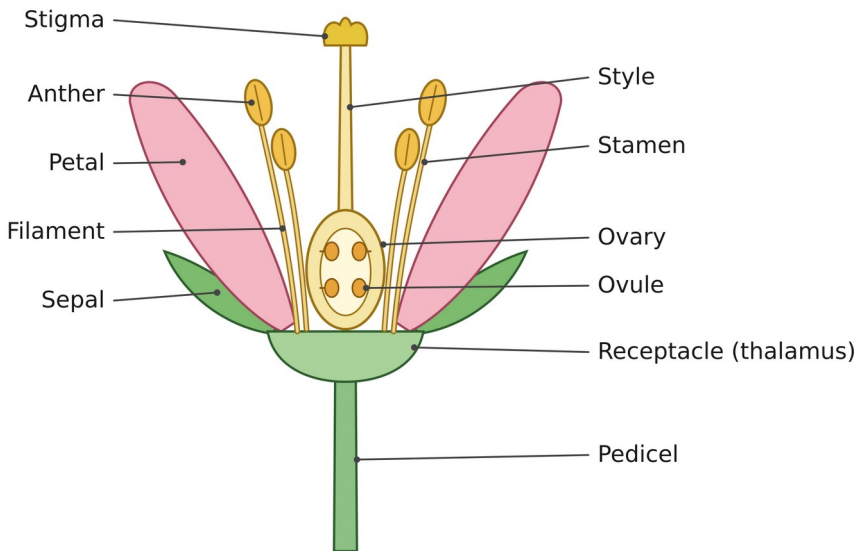


Fig. 8: Longitudinal section of a flower

Stamen = Anther + Filament

- The **anther** produces **pollen grains**, which are usually yellowish. Pollen grains contain the **male germ cells**.
- The **filament** is the stalk that holds the anther up.

Pistil = Stigma + Style + Ovary

- **Stigma** — the terminal part, often **sticky**, which receives pollen.
- **Style** — the **elongated middle part**; the pollen tube grows through it.
- **Ovary** — the **swollen bottom part**; it contains **ovules**, and each ovule has an **egg cell (female germ cell)**.

> **Exam Tip**
>
> 'Draw a labelled diagram of the longitudinal section of a flower' is an NCERT exercise question and appears often in board papers. Label at least: sepal, petal, anther, filament, stigma, style, ovary, ovule and receptacle.

6.2 Unisexual and bisexual flowers

Point	Unisexual flowers	Bisexual flowers
Reproductive parts	**Either** stamens **or** pistil	**Both** stamens and pistil
Flowers	Separate male and female flowers	Male and female parts in the same flower
Examples	Papaya, watermelon	*Hibiscus*, mustard

6.3 Pollination

Pollination is the transfer of **pollen grains from the anther to the stigma**. Agents of pollination include **wind, water, insects, birds and other animals**.

- **Self-pollination:** pollen reaches the stigma of the **same flower** (or of another flower on the same plant).
- **Cross-pollination:** pollen reaches the stigma of a flower on **another plant of the same species**.

> **Note for NCERT answers**
>
> NCERT puts it simply: transfer of pollen within the same flower is self-pollination, and transfer from one flower to another is cross-pollination. Strictly speaking, cross-pollination means transfer to a flower on a different plant of the same species.

Point	Self-pollination	Cross-pollination
Pollen goes to	Stigma of the same flower (or the same plant)	Stigma of a flower on another plant of the same species
Agent needed	Usually not	Yes — wind, water, insects, birds
Variation in offspring	Little	More
Examples	Pea, wheat	Papaya, maize

6.4 Fertilisation in plants

- After a pollen grain lands on a **suitable stigma**, its male germ cell has to reach the female germ cell in the ovule.
- A **pollen tube** grows out of the pollen grain, travels **through the style** and reaches the **ovule** in the ovary.
- The male germ cell moves down the tube and **fuses with the egg** in the ovule. This fusion is **fertilisation**, and the fertilised egg is called the **zygote**.

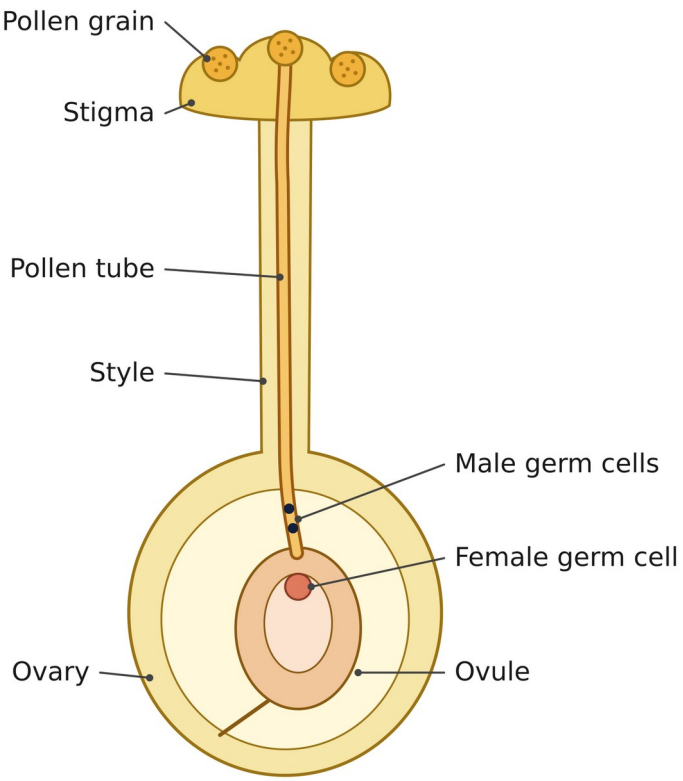


Fig. 9: Germination of pollen on the stigma

Point	Pollination	Fertilisation
What happens	Pollen is transferred from anther to stigma	Male and female germ cells fuse
Where	On the stigma (outside the ovary)	Inside the ovule (in the ovary)
How	By agents such as wind, water and insects	A pollen tube carries the male germ cell to the ovule
Order	Comes first	Follows pollination
Result	Pollen lands on the stigma and germinates	A zygote is formed

6.5 Changes after fertilisation

Part	What it becomes
Zygote	Divides many times to form the **embryo** inside the ovule
Ovule	Develops a tough coat and becomes the **seed**
Ovary	Grows rapidly and ripens to become the **fruit**
Petals, sepals, stamens, style, stigma	Usually shrivel and fall off (in some fruits, such as brinjal and tomato, the sepals remain)

> **Remember**
>
> **Ovule → Seed** and **Ovary → Fruit**. Trick: the smaller part (ovule) becomes the smaller structure (seed); the bigger part (ovary) becomes the bigger structure (fruit) that holds the seeds.

6.6 Seed and germination

- The seed contains the **future plant — the embryo**. Under suitable conditions, the embryo grows into a **seedling**; this process is **germination**.
- Parts of a dicot seed such as **Bengal gram (chana)**:
  - **Seed coat** — the protective outer covering;
  - **Cotyledons** — two seed leaves that **store food** for the embryo;
  - **Plumule** — grows into the **shoot**;
  - **Radicle** — grows into the **root**.
- Germination needs **water, air (oxygen) and a suitable temperature**.

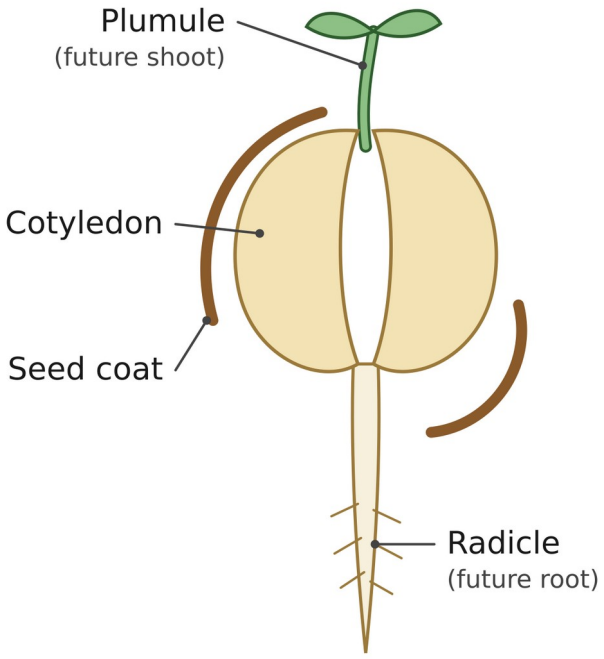
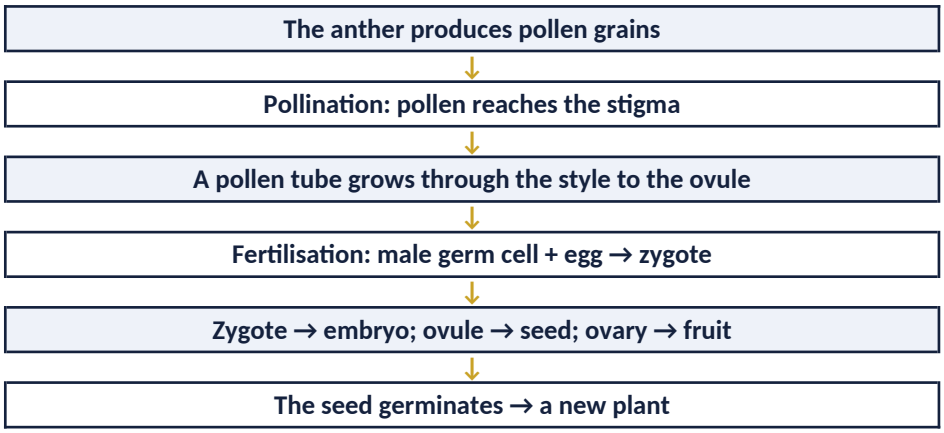


Fig. 10: Germination of a gram seed

NCERT Activity — Parts of a seed

Soak a few Bengal gram seeds overnight. Drain the water, cover the seeds with a wet cloth and leave them for a day (do not let them dry). Carefully cut open the seeds and identify the seed coat, cotyledons, plumule and radicle.

6.7 The whole sequence at a glance



7. Reproduction in Human Beings

7.1 Puberty — sexual maturation

From birth, the body keeps growing in size. During **adolescence**, general body growth slows down and the **reproductive tissues begin to mature**. This period of adolescence is called **puberty**.

Changes in both boys and girls	Changes in girls	Changes in boys
Thick hair grows in the armpits and in the genital area between the thighs	Breast size begins to increase; the skin of the nipples darkens	Thick hair begins to grow on the face (moustache, beard)
Thinner hair appears on the legs, arms and face	Menstruation begins	The voice begins to crack and becomes deeper
The skin becomes oily; pimples may appear		The penis occasionally becomes enlarged and erect, in daydreams or at night
Greater awareness of one's own body and of others' bodies		

- These changes happen **slowly, over months and years**. They do not all happen at once, and they do not happen at the same age in everyone.
- **Why do these changes happen?** They signal **sexual maturation** — the body is being prepared for reproduction:
  - in sexual reproduction, the germ cells of two individuals must meet, so male germ cells have to be transferred into the female body;
  - the woman's body must be able to **carry the baby** in the uterus for about nine months and **feed it (breastfeeding)** after birth.
- The changes are brought about by hormones: **testosterone** in boys and **oestrogen** in girls (link: Chapter 6, Control and Coordination).

7.2 Male reproductive system

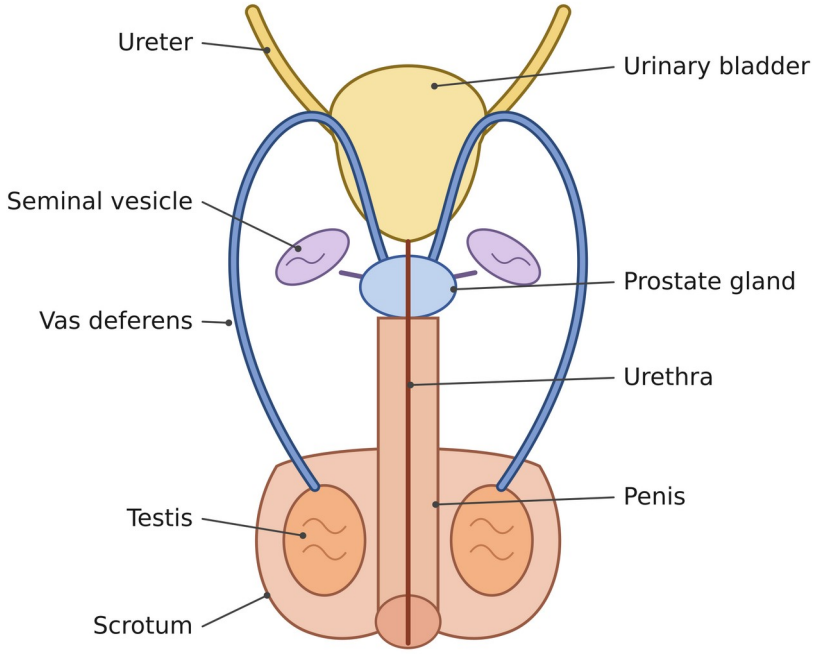


Fig. 11: Human male reproductive system

Organ	Structure and function
Testes (pair)	The primary male reproductive organs. They lie **outside the abdominal cavity, in the scrotum**, because **sperm formation needs a temperature lower than normal body temperature** (about 2–3 °C lower). They (i) **produce sperms** and (ii) secrete the hormone **testosterone**, which regulates sperm formation and brings about the changes seen in boys at puberty.
Scrotum	A pouch of skin that holds the testes outside the body cavity, keeping them cooler.
Vas deferens	The tube that carries sperms from the testis. It **joins with a tube coming from the urinary bladder**.
Urethra	The **common passage for both sperms and urine**; it runs through the penis.
Seminal vesicles and prostate gland	Add their secretions to the sperms, so that the sperms are in a **fluid**. This makes their **transport easier** and also provides **nutrition**. Sperms + these secretions = **semen**.
Penis	Transfers the sperms into the female reproductive tract (vagina) during sexual intercourse.

- **Sperms** are tiny bodies made mostly of **genetic material**, with a **long tail** that helps them swim towards the female germ cell.

Testis → **Vas deferens** → **Urethra** → **Penis**

7.3 Female reproductive system

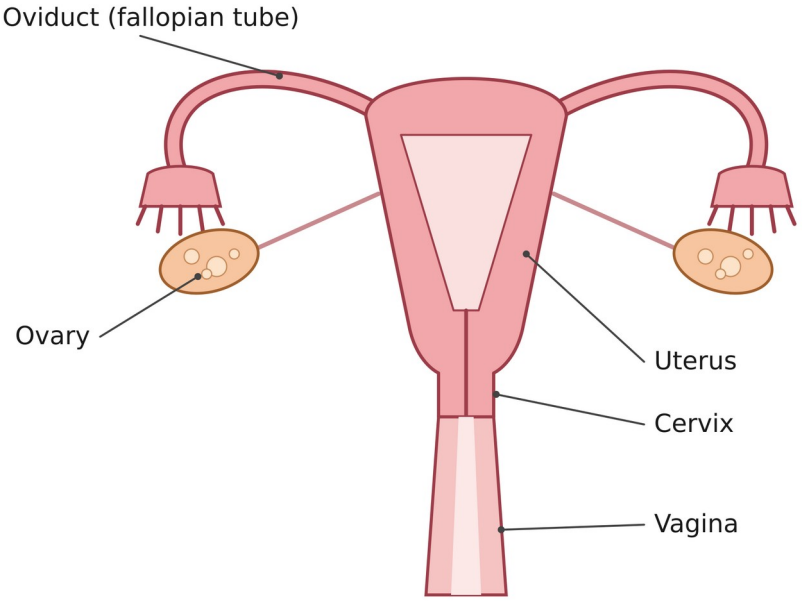
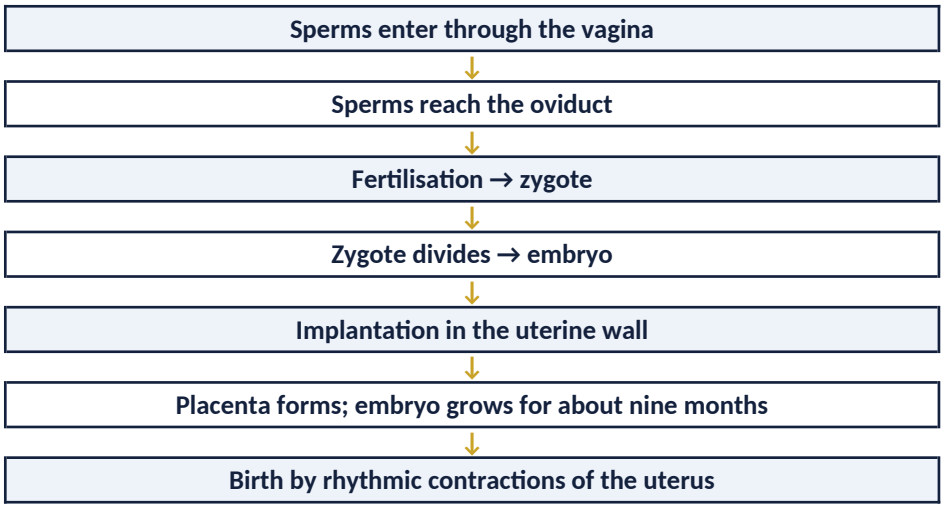


Fig. 12: Human female reproductive system

Organ	Structure and function
Ovaries (pair)	The primary female reproductive organs. They **produce eggs (ova)** and secrete the hormones **oestrogen and progesterone**. A girl is born with **thousands of immature eggs**; from puberty some begin to mature, and **one egg is released every month** by one of the ovaries.
Oviducts (fallopian tubes)	Thin tubes that carry the egg from the ovary to the uterus. **Fertilisation takes place here.**
Uterus (womb)	The two oviducts unite into an **elastic, bag-like, muscular** organ. The embryo is implanted and develops here.
Cervix	The narrow lower end of the uterus, which opens into the vagina.
Vagina	Receives the sperms during sexual intercourse; also serves as the **birth canal**.

7.4 Fertilisation, implantation and development

1. During sexual intercourse, sperms enter through the **vaginal passage**.
2. They travel upwards through the cervix and uterus and reach the **oviduct**.
3. If they meet an egg there, **fertilisation** takes place and a **zygote** (fertilised egg) is formed.
4. The zygote starts **dividing** and becomes a ball of cells — the **embryo** — while it moves down to the uterus.
5. The embryo is **implanted** in the thick, soft lining of the **uterus**. (Every month the uterus prepares for this: its lining becomes thick and well supplied with blood.)
6. The embryo develops further, forming tissues and organs, and is nourished through the **placenta**.
7. Development inside the mother's body takes about **nine months** (the **gestation period**).
8. The baby is born as a result of **rhythmic contractions of the muscles of the uterus**.

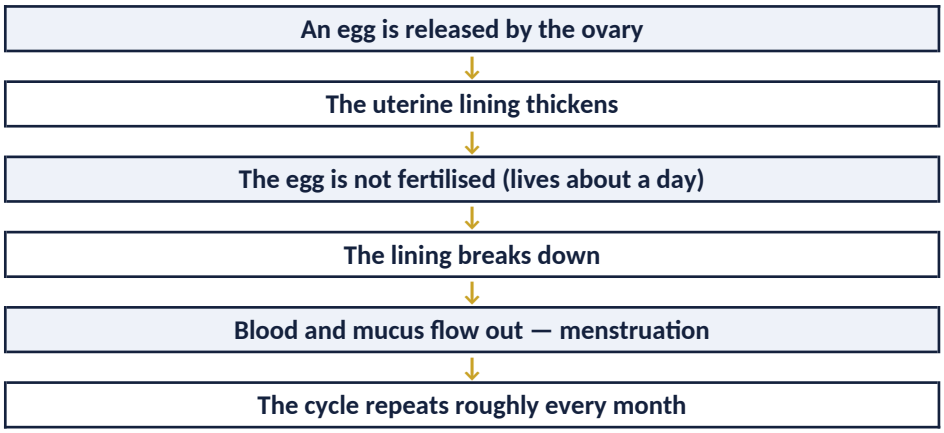


The placenta — how the embryo is nourished

- The embryo gets its nutrition from the **mother's blood** through a special tissue called the **placenta**.
- The placenta is a **disc embedded in the uterine wall**.
- On the embryo's side it has finger-like projections called **villi**; on the mother's side there are **blood spaces** that surround the villi.
- The villi give a **large surface area** for **glucose and oxygen** to pass from the mother's blood to the embryo.
- **Wastes** produced by the embryo pass into the **mother's blood** through the placenta and are removed.

7.5 What happens when the egg is not fertilised? — Menstruation

- Every month the ovary releases **one egg**, and the uterus prepares for a possible pregnancy: its lining becomes **thick and spongy**, with a rich blood supply, ready to nourish an embryo.
- If the egg is **not fertilised**, it lives for about **one day**.
- The thick lining is then not needed. It **breaks down slowly** and comes out through the vagina as **blood and mucus**. This is **menstruation**.
- The cycle repeats **roughly every month** (about 28 days), and the bleeding usually lasts **about 2 to 8 days**.
- The first menstruation at puberty is called **menarche**; the natural stopping of menstrual cycles later in life (around 45–50 years) is called **menopause**.



8. Reproductive Health

8.1 Making responsible choices

- Sexual maturation is **gradual**, and it happens while general body growth is still going on. Some degree of sexual maturity does **not** mean that the body or the mind is ready for sexual activity, or for having and bringing up children.
- Young people may feel **pressure** about such matters — from friends, and sometimes from family or society. Correct information helps them make **responsible decisions**.
- **Reproductive health** means well-being in all aspects of reproduction — physical, emotional, behavioural and social.

8.2 Sexually transmitted diseases (STDs)

Because the sexual act is a very close contact between two bodies, many infections can pass from one person to another this way.

Type	Examples
Bacterial	Gonorrhoea, syphilis
Viral	Warts, HIV-AIDS

- Using a **condom** (a covering for the penis) during sex helps to prevent the spread of many of these infections to some extent.

8.3 Safe sex vs HIV/AIDS

- **AIDS** (Acquired Immuno Deficiency Syndrome) is caused by **HIV** (Human Immunodeficiency Virus). HIV damages the body's **immune system**, so the person easily catches other infections.
- **HIV spreads through:**
  - unprotected sexual contact with an infected person;
  - transfusion of infected blood;
  - sharing of infected needles and syringes;
  - an infected mother to her baby (during pregnancy, birth or breastfeeding).
- **HIV does not spread by** touching, hugging, shaking hands, sharing food or utensils, or mosquito bites.
- **Safe practices:**
  - use of condoms, and a mutually faithful relationship with an uninfected partner;
  - only **screened (tested) blood** for transfusions;
  - **disposable or properly sterilised** needles and syringes;
  - testing and treatment during pregnancy to protect the baby.
- There is **no vaccine and no complete cure** for AIDS yet; medicines (antiretroviral therapy) can only keep the virus under control. So **prevention** is the best protection.

8.4 Contraception — preventing pregnancy

Why contraception?

- The sexual act always has the potential to lead to **pregnancy**.
- Pregnancy makes major demands on a woman's **body and mind**. If she is not ready for it, her **health is adversely affected**.

- Reasons for adopting contraceptive methods:
  - to avoid **unwanted pregnancies**;
  - to keep a proper **gap (spacing)** between children, so that the mother recovers and each child gets proper care;
  - to protect the **health of the woman** — frequent pregnancies weaken her body;
  - to keep the **family size** in line with its resources and to check **population growth**;
  - some methods (condoms) also help prevent **STDs**, including HIV-AIDS.

Methods of contraception

Method	How it works	Examples	Points to note
Barrier (mechanical)	A physical barrier stops sperms from reaching the egg	Condom for the penis; similar coverings worn in the vagina (diaphragm, cervical cap)	Condoms also give some protection against STDs
Hormonal (oral pills)	Changes the hormonal balance so that **eggs are not released**; fertilisation cannot occur	Oral contraceptive pills	Can cause side-effects, because they alter the hormonal balance
Intrauterine devices	A device placed in the uterus prevents pregnancy	Loop, copper-T	May cause side-effects due to irritation of the uterus; **no protection against STDs**
Surgical (sterilisation)	The path of the gametes is blocked	**Vasectomy** — vas deferens blocked in males, so sperms cannot be transferred; **tubectomy** — fallopian tube blocked in females, so the egg cannot reach the uterus	A permanent method; surgery can cause infection and other problems if not done properly

> **Copper-T and STDs (NCERT in-text question)**
>
> A woman using a copper-T is **not** protected from sexually transmitted diseases. The copper-T only helps prevent pregnancy; it does not stop the exchange of body fluids during sex. Protection against STDs needs a barrier method such as a condom.

8.5 Termination of pregnancy and female foeticide

- Surgical methods can also be used to **remove unwanted pregnancies**. These can be **misused** by people who do not want a particular child — as in the **illegal sex-selective abortion of female foetuses (female foeticide)**.
- For a healthy society, the **female–male sex ratio** must be maintained. Because of reckless female foeticide, the **child sex ratio** has been falling at an alarming rate in some sections of society.
- **Prenatal sex determination is prohibited by law** in India (the PC-PNDT Act).

8.6 Reproduction and population

- The size of a population depends on its **birth rate and death rate**.
- A very large population makes it **harder to improve everyone's standard of living**.
- However, if **inequality in society** is the main reason for a poor standard of living for many people, population size is relatively less important.

8.7 Child bearing and women's health

- The legal minimum age of marriage in India is **18 years for women and 21 years for men** (Prohibition of Child Marriage Act, 2006).
- Pregnancy before the body is fully mature is risky for both mother and baby (for example, anaemia, a difficult delivery, low birth weight).
- A woman's health is protected by:
  - having children only when her **body and mind are ready**;
  - a gap of at least **two years** between children;
  - a **nutritious diet** (rich in iron, calcium and folic acid) during pregnancy;
  - regular **antenatal check-ups**, the recommended vaccinations, and delivery by trained health workers;
  - an **equal say** in decisions about the size of the family.
- A healthy mother means a healthy child — and a healthier society.

9. Quick Revision

9.1 Key terms

Term	Meaning
Reproduction	Production of new individuals of the same kind; ensures continuity of the species
DNA	Molecule in the chromosomes that carries the information for making proteins and decides body design
Variation	Differences among individuals of a species, from imperfect DNA copying and (in sexual reproduction) from combining the DNA of two parents
Clone	An offspring that is genetically identical to its parent (asexual reproduction)
Binary fission	Division of a parent cell into two equal daughter cells
Multiple fission	Division of a parent cell into many daughter cells at the same time
Fragmentation	Breaking of a mature body into pieces, each growing into a new organism
Regeneration	Formation of a whole organism from a body part, by proliferation and differentiation of specialised cells
Budding	Formation of a new individual from an outgrowth (bud) of the parent
Vegetative propagation	Growing new plants from a root, stem or leaf
Tissue culture	Growing new plants from cells or tissue placed in an artificial medium
Callus	Mass of undifferentiated cells formed in tissue culture
Hyphae	Thread-like structures that make up the body of *Rhizopus*
Sporangium	Blob-like structure that produces and holds spores
Spore	Small reproductive cell with a thick protective wall
Gamete (germ cell)	Reproductive cell with half the number of chromosomes — the sperm or the egg
Fertilisation	Fusion of a male gamete with a female gamete
Zygote	The cell formed by fertilisation
Pollination	Transfer of pollen grains from the anther to the stigma
Pollen tube	Tube that grows from a pollen grain through the style to the ovule

Term	Meaning
Ovule	Structure in the ovary that contains the egg; it becomes the seed
Embryo	Early stage of development formed from the zygote
Plumule / Radicle	Parts of the embryo that grow into the shoot / the root
Cotyledon	Seed leaf that stores food
Germination	Growth of the embryo in a seed into a seedling
Puberty	Period of adolescence when the reproductive tissues mature
Scrotum	Pouch that holds the testes outside the abdominal cavity
Vas deferens	Tube that carries sperms from the testis
Semen	Sperms together with the secretions of the seminal vesicles and prostate gland
Oviduct (fallopian tube)	Tube that carries the egg from the ovary to the uterus; the site of fertilisation
Implantation	Embedding of the embryo in the lining of the uterus
Placenta	Disc-shaped tissue that links the embryo to the uterine wall for exchange of food, oxygen and wastes
Gestation	Development of the baby inside the mother's body (about nine months)
Menstruation	Monthly breakdown and discharge of the uterine lining when the egg is not fertilised
Contraception	Methods used to prevent pregnancy
Vasectomy / Tubectomy	Surgical blocking of the vas deferens / the fallopian tube
STD	Sexually transmitted disease — e.g., gonorrhoea, syphilis, warts, HIV-AIDS

9.2 Organism and its mode of reproduction

Organism	Mode of reproduction
Amoeba	Binary fission (in any plane); multiple fission inside a cyst
Leishmania	Binary fission in a definite plane
Paramecium**, bacteria**	Binary fission
Plasmodium	Multiple fission
Spirogyra	Fragmentation
Planaria	Regeneration
Hydra	Budding (also regeneration)
Yeast	Budding
Rhizopus	Spore formation
Bryophyllum	Vegetative propagation by buds on the leaf margin
Potato	Vegetative propagation by the stem tuber ('eyes')
Banana, rose, jasmine	Vegetative propagation
Flowering plants, human beings	Sexual reproduction

9.3 Diagrams to practise

Diagram	Labels to show
Binary fission in *Amoeba*	Nucleus, cytoplasm, dividing nucleus, daughter cells
Multiple fission in *Plasmodium*	Parent cell, many nuclei, daughter cells
Budding in *Hydra*	Parent *Hydra*, bud, tentacles, new *Hydra*
Regeneration in *Planaria*	Cut pieces, each regenerating a whole body
Spore formation in *Rhizopus*	Hyphae, sporangium, spores
LS of a flower (very important)	Sepal, petal, anther, filament, stigma, style, ovary, ovule, receptacle
Germination of pollen on stigma (very important)	Pollen grain, stigma, pollen tube, style, ovary, ovule, female germ cell, male germ cell
Germination of a seed	Seed coat, cotyledon, plumule, radicle
Male reproductive system (very important)	Urinary bladder, ureter, seminal vesicle, prostate gland, vas deferens, urethra, penis, testis, scrotum
Female reproductive system (very important)	Ovary, oviduct (fallopian tube), uterus, cervix, vagina

9.4 Memory tricks

Mnemonics

- **Asexual modes — "Funny Frogs Rarely Buy Vegetable Soup":** Fission, Fragmentation, Regeneration, Budding, Vegetative propagation, Spore formation.
- **Flower parts, outside to inside — "Some People Sing Proudly":** Sepal, Petal, Stamen, Pistil.
- **Pistil, top to bottom — "Students Study Often":** Stigma, Style, Ovary.
- **Path of sperms — "Tired Vans Use Petrol":** Testis → Vas deferens → Urethra → Penis.
- **Female tract, ovary downwards — "Our Old Uncle Can't Vanish":** Ovary → Oviduct → Uterus → Cervix → Vagina.

9.5 Common mistakes to avoid

- Fertilisation in humans takes place in the **oviduct** — not in the uterus or the ovary.
- **Pollination** (transfer of pollen) is not the same as **fertilisation** (fusion of gametes).
- **Ovule → seed** and **ovary → fruit** — not the other way round.
- The testes lie in the scrotum because sperm formation needs a **lower temperature** — not "for protection".
- A copper-T does **not** protect against STDs.
- *Amoeba* divides in **any** plane; *Leishmania* divides in a **definite** plane.
- Regeneration is **not** a normal method of reproduction for most organisms.
- Spell the names correctly: *Amoeba*, *Leishmania*, *Plasmodium*, *Spirogyra*, *Planaria*, *Hydra*, *Rhizopus*, *Bryophyllum*.

10. NCERT Questions — Answered

10.1 In-text questions

Q1. What is the importance of DNA copying in reproduction?

Ans. DNA carries the information for making proteins, and these proteins decide the body design. DNA copying makes sure that each new cell — and so each offspring — receives this full set of information, so the offspring have the same basic body design as the parent. Because copying is not perfectly accurate, it also introduces slight variations, which help the species survive changes over time.

Q2. Why is variation beneficial to the species but not necessarily to the individual?

Ans. If the niche of a population changes drastically (for example, the water temperature rises), individuals with suitable variations may survive while others die, so the species continues and can evolve. For a particular individual, however, a variation may be useless or even harmful. Its value shows up only at the population level, when conditions change.

Q3. How does binary fission differ from multiple fission?

Ans. In binary fission, the nucleus divides once and the parent cell divides into **two** daughter cells (e.g., *Amoeba*, *Leishmania*). In multiple fission, the nucleus divides repeatedly and the parent cell divides into **many** daughter cells at the same time (e.g., *Plasmodium*).

Q4. How will an organism benefit if it reproduces through spores?

Ans.

- Spores are produced in very large numbers.
- They are tiny and light, so air carries them far, and the organism spreads to new places.
- Their thick walls protect them in unfavourable conditions; they germinate when they reach a moist surface.

Q5. Why can't more complex organisms give rise to new individuals through regeneration?

Ans. Complex organisms have highly specialised cells organised into tissues, organs and organ systems that depend on one another. Most of these cells have lost the ability to divide and form all the other types of cells, and there are not enough regenerative cells to rebuild an entire body from a part. So a body part cannot develop into a whole organism.

Q6. Why is vegetative propagation practised for growing some types of plants?

Ans.

- Plants such as banana, orange, rose and jasmine have lost the capacity to produce seeds, so they can be grown only this way.
- Plants raised this way flower and bear fruit earlier than plants grown from seeds.
- The new plants are genetically similar to the parent, so desirable features are preserved.
- It is quick, easy and economical.

Q7. Why is DNA copying an essential part of the process of reproduction?

Ans. The information for the body design of an organism is stored in its DNA. When a cell reproduces, each new cell must receive a copy of the DNA (along with cellular apparatus) to make the right proteins and work properly. Without DNA copying, the new individual would not get the blueprint of the parent's body design. DNA copying also brings small variations that are important for the survival of the species.

Q8. How is the process of pollination different from fertilisation?

Ans. Pollination is the transfer of pollen grains from the anther to the stigma; it takes place outside the ovary with the help of agents such as wind, water and insects. Fertilisation is the fusion of the male germ cell with the female germ cell (egg) inside the ovule, after a pollen tube carries the male germ cell to the ovule. Pollination comes first; fertilisation produces the zygote.

Q9. What is the role of the seminal vesicles and the prostate gland?

Ans. They add their secretions to the sperms. These secretions provide a fluid medium (semen) in which the sperms are transported easily, and they also provide nutrition to the sperms.

Q10. What are the changes seen in girls at the time of puberty?

Ans.

- The breasts begin to develop and the skin of the nipples darkens.
- Menstruation begins.
- Thick hair grows in the armpits and the genital area; thinner hair appears on the arms and legs.
- The skin becomes oily and pimples may appear.

Q11. How does the embryo get nourishment inside the mother's body?

Ans. Through the placenta, a disc-like tissue embedded in the uterine wall. Its villi on the embryo's side are surrounded by blood spaces on the mother's side. The villi give a large surface area through which glucose and oxygen pass from the mother's blood to the embryo; wastes from the embryo pass into the mother's blood.

Q12. If a woman is using a copper-T, will it help in protecting her from sexually transmitted diseases?

Ans. No. A copper-T is placed in the uterus to prevent pregnancy, but it does not prevent contact of body fluids during sex, so it cannot protect her from STDs. A barrier method such as a condom is needed for that.

10.2 Exercise questions

Q1. Asexual reproduction takes place through budding in

(a) *Amoeba* (b) yeast (c) *Plasmodium* (d) *Leishmania*

Ans. (b) yeast

Q2. Which of the following is not a part of the female reproductive system in human beings?

(a) ovary (b) uterus (c) vas deferens (d) fallopian tube

Ans. (c) vas deferens — it is part of the male reproductive system.

Q3. The anther contains

(a) sepals (b) ovules (c) carpel (d) pollen grains

Ans. (d) pollen grains

Q4. What are the advantages of sexual reproduction over asexual reproduction?

Ans.

- The DNA of two parents combines, so the offspring show more variation. (In asexual reproduction, variation comes only from small DNA-copying errors.)
- Variation helps the species adapt and survive in a changing environment.
- It provides the raw material for evolution.
- Useful features of two parents can come together in the offspring.

Q5. What are the functions performed by the testis in human beings?

Ans. (i) It produces the male germ cells — the sperms. (ii) It secretes the hormone testosterone, which regulates the formation of sperms and brings about the changes in boys at puberty (such as facial hair and a deeper voice).

Q6. Why does menstruation occur?

Ans. Every month the ovary releases an egg, and the uterus prepares to receive a fertilised egg — its lining becomes thick and spongy, with a rich blood supply. If the egg is not fertilised, this lining is not needed. It breaks down and is discharged through the vagina as blood and mucus. This is menstruation; it occurs roughly every month and lasts about 2 to 8 days.

Q7. Draw a labelled diagram of the longitudinal section of a flower.

Ans. See Fig. 8 in Section 6.1. Labels: sepal, petal, anther, filament, stigma, style, ovary, ovule and receptacle.

Q8. What are the different methods of contraception?

Ans.

- **Barrier methods** — condoms and similar coverings worn in the vagina stop sperms from reaching the egg.
- **Hormonal methods** — oral pills change the hormonal balance so that eggs are not released.
- **Intrauterine devices** — a loop or copper-T placed in the uterus prevents pregnancy.
- **Surgical methods** — vasectomy (vas deferens blocked in males) and tubectomy (fallopian tube blocked in females).

Q9. How are the modes of reproduction different in unicellular and multicellular organisms?

Ans. In unicellular organisms the single cell carries out all life processes, so reproduction is simply cell division — binary fission (*Amoeba*), multiple fission (*Plasmodium*) or budding (yeast). Multicellular organisms have specialised cells, tissues and organs, so they cannot reproduce by simple cell division. Simple multicellular organisms use fragmentation (*Spirogyra*), regeneration (*Planaria*), budding (*Hydra*), spore formation (*Rhizopus*) or vegetative propagation; complex ones reproduce sexually, using specialised reproductive organs that produce gametes.

Q10. How does reproduction help in providing stability to populations of species?

Ans. Reproduction produces new individuals to replace those that die, so a species does not disappear. Consistent DNA copying keeps the body design suited to the species' niche, while the small variations that arise allow some individuals to survive if the niche changes. Together, these keep populations stable over generations.

Q11. What could be the reasons for adopting contraceptive methods?

Ans.

- To avoid unwanted pregnancies.
- To keep a proper gap between children.
- To protect the health of the woman, since frequent pregnancies harm her body and mind.
- To keep the family size in line with its resources and to control population growth.
- To prevent the spread of STDs (condoms).

11. Exam Practice

11.1 Multiple choice questions (1 mark each)

Q1. Binary fission in a definite plane is seen in
(a) *Leishmania* (b) *Amoeba* (c) *Plasmodium* (d) yeast

Q2. Multiple fission occurs in
(a) *Paramecium* (b) *Hydra* (c) *Plasmodium* (d) *Planaria*

Q3. *Spirogyra* reproduces asexually by
(a) budding (b) spore formation (c) regeneration (d) fragmentation

Q4. The 'blob-on-a-stick' structures seen on bread mould are
(a) hyphae (b) sporangia (c) spores (d) buds

Q5. New plants growing from the buds in the notches of a *Bryophyllum* leaf is an example of
(a) budding (b) fragmentation (c) vegetative propagation (d) regeneration

Q6. In tissue culture, the mass of undifferentiated cells is called
(a) callus (b) clone (c) embryo (d) plantlet

Q7. After fertilisation, the ovary develops into the
(a) seed (b) seed coat (c) embryo (d) fruit

Q8. The pollen tube grows through the
(a) filament (b) style (c) sepal (d) petal

Q9. The part of the embryo that grows into the shoot is the
(a) radicle (b) cotyledon (c) plumule (d) seed coat

Q10. Which of the following has unisexual flowers?
(a) *Hibiscus* (b) mustard (c) rose (d) papaya

Q11. In human females, fertilisation normally takes place in the
(a) oviduct (b) uterus (c) ovary (d) vagina

Q12. The testes lie in the scrotum because
(a) there is more space (b) it protects them from injury (c) sperm formation needs a lower temperature (d) it helps in passing urine

Q13. The common passage for urine and sperms in human males is the
(a) vas deferens (b) urethra (c) ureter (d) seminal vesicle

Q14. Which contraceptive method also gives protection against STDs?
(a) copper-T (b) oral pills (c) vasectomy (d) condom

Q15. Which of the following STDs is caused by a virus?
(a) HIV-AIDS (b) syphilis (c) gonorrhoea (d) all of these

Q16. The number of chromosomes in a human egg cell is
(a) 46 (b) 44 (c) 23 (d) 22

11.2 Assertion–Reason questions (1 mark each)

> **Directions**
>
> Choose: **(a)** Both A and R are true, and R is the correct explanation of A. **(b)** Both A and R are true, but R is not the correct explanation of A. **(c)** A is true, but R is false. **(d)** A is false, but R is true.

Q1. **Assertion (A):** Variation is beneficial for the survival of a species. **Reason (R):** Some variants may survive when the environment changes drastically.

Q2. **Assertion (A):** The testes are located outside the abdominal cavity. **Reason (R):** Sperm formation requires a temperature lower than normal body temperature.

Q3. **Assertion (A):** A copper-T protects a woman from sexually transmitted diseases. **Reason (R):** A copper-T is placed in the uterus to prevent pregnancy.

Q4. **Assertion (A):** Complex multicellular organisms cannot reproduce by fragmentation. **Reason (R):** Their cells are organised into specialised tissues and organs.

Q5. **Assertion (A):** Plants raised by vegetative propagation show a lot of variation. **Reason (R):** They are produced from a single parent without the fusion of gametes.

Q6. **Assertion (A):** Reproduction is essential for the survival of an individual organism. **Reason (R):** Reproduction ensures the continuity of a species.

Q7. **Assertion (A):** The ovule develops into a seed after fertilisation. **Reason (R):** The ovary ripens to form the fruit.

Q8. **Assertion (A):** Gametes contain half the number of chromosomes of body cells. **Reason (R):** This keeps the chromosome number constant from one generation to the next.

11.3 Answer key — MCQs and Assertion–Reason

MCQ	1 (a)	2 (c)	3 (d)	4 (b)	5 (c)	6 (a)	7 (d)	8 (b)
MCQ	9 (c)	10 (d)	11 (a)	12 (c)	13 (b)	14 (d)	15 (a)	16 (c)
A–R	1 (a)	2 (a)	3 (d)	4 (a)	5 (d)	6 (d)	7 (b)	8 (a)

11.4 Short answer questions (2–3 marks)

Q1. Why is pollination essential for fertilisation in flowering plants?

Ans. The male germ cells are inside the pollen grains. Only when pollen reaches the stigma can a pollen tube grow through the style to the ovule and deliver the male germ cell to the egg. Without pollination, the male and female germ cells cannot meet.

Q2. Name the three parts of the pistil and give one function of each.

Ans. **Stigma** — receives the pollen. **Style** — the pollen tube grows through it to reach the ovary. **Ovary** — contains the ovules with egg cells, and develops into the fruit after fertilisation.

Q3. Give two advantages and one disadvantage of tissue culture.

Ans. Advantages: a large number of plants can be raised from one parent in a short time; the plants are disease-free. Disadvantage: all the plants are genetically identical, so one disease can destroy all of them.

Q4. What is the placenta? State two of its functions.

Ans. The placenta is a disc-like tissue embedded in the uterine wall that connects the embryo with the mother. (i) It passes glucose and oxygen from the mother's blood to the embryo. (ii) It passes the wastes of the embryo into the mother's blood for removal.

Q5. Give two ways in which HIV spreads and two ways of preventing it.

Ans. Spread: unprotected sexual contact with an infected person; transfusion of infected blood or sharing of infected needles (also from an infected mother to her baby). Prevention: use of condoms; using only screened blood and disposable or sterilised needles.

Q6. Why does the lining of the uterus become thick every month? What happens if fertilisation does not take place?

Ans. The lining becomes thick, spongy and rich in blood so that it can receive and nourish an embryo if fertilisation occurs. If fertilisation does not take place, the egg lives for about a day; the lining is not needed, so it breaks down and comes out as blood and mucus — menstruation.

Q7. Why should female foeticide be prevented?

Ans. It is illegal and unethical. It disturbs the female–male sex ratio, which must be maintained for a healthy society; the child sex ratio is already falling at an alarming rate in some sections of society.

11.5 Long answer questions (5 marks)

Q1. (a) Draw a labelled diagram of the longitudinal section of a flower. (b) Describe the events that take place from pollination to the formation of the seed and the fruit.

Ans. (a) See Fig. 8. (b) Pollen grains are transferred from the anther to the stigma (pollination). On a suitable stigma, the pollen grain germinates and a pollen tube grows through the style to reach the ovule in the ovary. The male germ cell travels down the tube and fuses with the egg in the ovule (fertilisation), forming a zygote. The zygote divides repeatedly to form the embryo. The ovule develops a tough coat and becomes the seed, and the ovary grows and ripens into the fruit. The petals, sepals, stamens, style and stigma usually shrivel and fall off.

Q2. (a) Draw a labelled diagram of the human male reproductive system. (b) State the functions of the testes, the vas deferens, and the seminal vesicles and prostate gland.

Ans. (a) See Fig. 11. (b) **Testes:** produce sperms and secrete testosterone (they lie in the scrotum because sperm formation needs a lower temperature). **Vas deferens:** carries sperms from the testis to the urethra. **Seminal vesicles and prostate gland:** add secretions that make the sperms flow in a fluid (semen), easing their transport and providing nutrition.

Q3. (a) Name the parts of the human female reproductive system and state the function of each. (b) What happens to the egg if (i) it is fertilised, (ii) it is not fertilised?

Ans. (a) **Ovaries** — produce eggs and the hormones oestrogen and progesterone. **Oviducts** — carry the egg to the uterus; fertilisation takes place here. **Uterus** — the embryo is implanted and develops here. **Cervix** — opens into the vagina. **Vagina** — receives sperms; it is also the birth canal. (b)(i) The zygote divides to form an embryo, which is implanted in the uterine wall, receives nourishment through the placenta and develops for about nine months until birth. (ii) The egg lives for about a day; the thickened uterine lining breaks down and is discharged as blood and mucus (menstruation).

Q4. What is contraception? Describe four methods of contraception, with one limitation of each.

Ans. Contraception is the use of methods to prevent pregnancy. **Barrier** (condom, diaphragm) — stops sperms from reaching the egg; limitation: must be used correctly every time. **Oral pills** — change the hormonal balance so that eggs are not released; limitation: may cause side-effects. **Copper-T / loop** —

placed in the uterus; limitation: may irritate the uterus and gives no protection against STDs. **Surgical** (vasectomy, tubectomy) — blocks the path of gametes; limitation: permanent, and surgery can cause infection if not done properly.

11.6 Case-based questions (4 marks each)

> **Case 1**
>
> A farmer near Motihari wants thousands of banana plants of a high-yielding variety, all giving fruit of the same quality. An agricultural centre supplies him with plantlets raised from the growing tip of a single healthy plant, grown in a nutrient medium under sterile conditions.

Q1. Name the technique used by the agricultural centre. (1)

Ans. Tissue culture (micropropagation).

Q2. What is the mass of undifferentiated cells formed in this technique called? (1)

Ans. Callus.

Q3. Give two advantages of this technique for the farmer. (1)

Ans. A large number of identical plants is obtained quickly; the plants are disease-free and give fruit of the same quality as the parent.

Q4. Why is it risky to grow only this one variety over a large area? (1)

Ans. All the plants are genetically identical (no variation), so a single disease or pest could destroy the entire crop.

> **Case 2**
>
> Riya kept a moist slice of bread in a dark cupboard. After a few days she saw a white cottony growth on it, with tiny black dots on top.

Q1. Name the organism growing on the bread. (1)

Ans. *Rhizopus* (bread mould).

Q2. What are the thread-like structures and the black dots called? (1)

Ans. The threads are hyphae; the black dots are sporangia, which contain spores.

Q3. Why did the growth appear on the moist bread and not on a dry slice? (1)

Ans. Spores germinate only when they come in contact with a moist surface; moisture and food in the bread support their growth.

Q4. State one advantage of this mode of reproduction. (1)

Ans. Very large numbers of light spores are produced and spread easily by air; their thick walls help them survive unfavourable conditions.